

 THE REALITY CHECK

AI interviews **don't** test definitions.

They test how you think under ambiguity, scale, latency, security, and production constraints.



Not: ~~"What is RAG?"~~



BUT THEY ASK:

"How do you debug retrieval failures & scale?"

7 question kits for AI engineers



</> DSA for AI Engineers

- > A process repeatedly retrieves the smallest/largest element from a changing collection. What structure is best?
- > Why would hashing improve performance compared to scanning when checking for value existence?
- > When parsing deeply nested JSON outputs from an LLM, what advantage does stack-based tracking provide?
- > In a streaming dataset, why is it inefficient to repeatedly recompute metrics over overlapping ranges?
- > Why might scanning forward repeatedly be inefficient to find the next larger value in an array?
- > Why does binary search fail to work correctly if embeddings are not sorted?
- > What approach compresses a scanned dataset by removing repeated consecutive entries in a single pass?
- > If one pointer moves twice as fast as another, what kinds of structural patterns could this detect?
- > Why is scanning a dataset repeatedly inefficient when counting token frequencies?
- > How do Trie data structures accelerate sub-word tokenization algorithms?

🔗 For more questions, check the caption for link.



GenAI Engineering

- › When should you switch from large models to smaller models for cost optimization?
- › How would you test GenAI systems against adversarial inputs?
- › Why can the CLS token represent the entire sequence in BERT-style models?
- › What is fine-tuning in the context of LLMs, and how does it differ from pretraining?
- › Why are transformer layers stacked repeatedly, and how does depth influence capability?
- › How would you combine vector search and structured databases for agent memory?
- › What challenges arise when processing extremely long text sequences?
- › Why might normalization improve model convergence?
- › How does underfitting uniquely manifest in generative models?
- › What are the practical tradeoffs between LoRA, QLoRA, and full fine-tuning?



For more questions, check the caption for link.

RAG

- How do you mitigate the "Lost in the Middle" phenomenon in LLMs?
- What is the optimal strategy for chunking hierarchical or tabular data?
- How do temperature, prompt structure, and token limits impact generation quality?
- What problem does LangGraph solve that traditional LangChain chains do not?
- How would you design session-level memory preventing data leakage across users?
- What techniques can reduce irrelevant or noisy retrieved context?
- What is a vector database, and why is ANN search required instead of exact search?
- How do you handle contradictory information retrieved from different documents?
- Explain how FLARE (Forward-Looking Active REtrieval) works.
- How do you evaluate the recall and precision of your retrieval pipeline independently?

 [For more questions, check the caption for link.](#)



AI Agents

- > How does an agent determine which tool to select among many?
- > How do you prevent runaway agents from exhausting API credits?
- > How would you design an enterprise tool registry compatible with MCP?
- > What are the risks of emergent behaviors in agent collaboration?
- > How does LangChain manage conversational memory, and what are its scaling limitations?
- > How do you measure the per-agent output validity rate when using JSON schemas?
- > What happens if an agent trusts incorrect or malicious tool outputs?
- > How do you evaluate if an agent's reasoning path was optimal?
- > How do you handle fallback mechanisms when a primary tool API is down?
- > What architectural patterns prevent infinite reasoning loops?



For more questions, check the caption for link.

- > What is the role of eval suites in production monitoring, and how do you run them on live traffic?
- > What is faithfulness in RAG evaluation? How do you measure if the response is grounded?
- > What is a cost-per-query budget and how do you enforce it in a production LLM system?
- > How do you use LLM traces to debug a failure that only occurs on specific user inputs?
- > What are the limitations of thumbs-up/thumbs-down feedback as a quality signal?
- > What is the difference between perplexity and task-based benchmark scores?
- > How do you evaluate a system where there is no single correct answer?
- > How do you detect and alert on data drift in your vector embeddings?
- > What strategies do you use for continuous integration (CI/CD) of prompts?
- > How do you manage model versioning and rollback in production?

 For more questions, check the caption for link.



AI System Design

- > You're reviewing a design where requirements are "accurate and fast." What's wrong with this, and how would you fix it?
- > A reporting feature generates long summaries; a chat feature needs immediate feedback. How do you design one system for both?
- > Design an agent system where downtime costs \$10,000/min. What reliability patterns apply?
- > Your AI serves 40 countries in 15 languages. How do you design multilingual retrieval reliably?
- > A user returns after two weeks. How does your system retrieve the right context without overloading the context window?
- > How do you implement semantic caching to reduce latency and API costs?
- > What is the architectural difference between stateless inference and stateful agents?
- > How do you handle load balancing for heavy LLM generation requests?
- > How would you design a scalable Vector DB architecture for 1 billion embeddings?
- > What is the cost tradeoff between running open-source models vs managed APIs at scale?



For more questions, check the caption for link.



AI Security

- > What happens if your LLM API key is exposed through logs?
- > Why is “the model is stateless” a misleading assumption when designing secure systems?
- > How would you design a system to prevent harmful actions from being executed based on LLM output?
- > What risks arise when you allow long conversation history in an AI system?
- > Why are multi-step agents more vulnerable than single-step systems?
- > What is instruction hierarchy, and how do you enforce it?
- > What adversarial tests would you prioritize for a public chatbot?
- > What happens if the model combines unrelated documents incorrectly?
- > How do you mitigate prompt injection in customer-facing applications?
- > How do you securely handle PII when sending data to third-party foundation models?



For more questions, check the caption for link.